



SCIENCE FOUNDATIONS

The Scientific Process

How scientists separate what is TRUE from what only sounds true.

PART 1

Being Skeptical



Everything You Hear Is Opinion



The Old Saying

"Believe nothing that you hear and only half of what you see."

Opinions and perspectives are useful — but they aren't facts. A scientist's job is to find what can be shown true, again and again.



What Scientists Chase

FACTS — things that can be shown true REPEATEDLY, by different people, in different labs, over time.

If a claim only holds up when one person tests it once, it's not yet a fact.

"A New Study Shows..."

Every year, headlines about coffee say:

 Coffee causes cancer

 Coffee prevents cancer

 Coffee harms your heart

 Coffee protects your heart

 Coffee shortens your life

 Coffee extends your life

So does coffee kill you, save you, or both? The answer lies in the SOURCES.

What Makes a Study Reliable?

1

Testable hypothesis

You can DO something to check it.

2

Well-designed experiment

The setup actually answers the question.

3

Controlled environment

Extra variables held constant.

4

Change one variable

Only ONE thing changes at a time.

5

Multiple trials

Repeat, repeat, repeat.

6

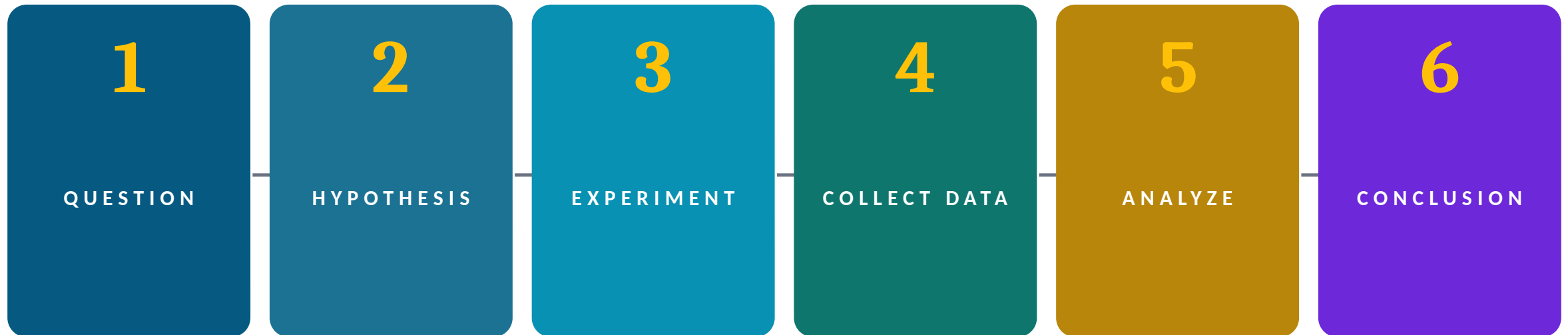
Remove researcher bias

Blinding, randomization, honest reporting.

PART 2

The Six Steps of the Scientific Process

The Six Steps



Repeatable. Reliable. Replicable.

Step 1 — Question



What Makes a Question Good

Pick something you're curious about — then check:

- Can you DO something to test it?
- Can you MEASURE the result?

Avoid vague words like 'better' or 'best' — those are opinions, not measurements.



Testable vs. Not

✓ Do fingers wrinkle FASTER in cold or hot water? (Timeable!)

✗ What tastes better — Oreos or Chips Ahoy? (Opinion — not measurable.)

Refining a Question

TOO VAGUE

Which is better, Tide or All?

Opinion. Not measurable.

BETTER

What laundry detergent works best on grass stains?

Closer, but 'best' is still fuzzy.

TESTABLE ✓

Which detergent — Tide or All — removes a grass stain with the least amount of detergent?

One IV, one DV, measurable.

Step 2 — Hypothesis

The Format

If DV is related to IV, then [prediction].



Example

If freezing time is related to starting water temperature, then cold water will freeze faster.



It's a Prediction, Not a Wish

A hypothesis doesn't have to be RIGHT — it just has to be a specific, testable guess about the relationship.

Step 3 — Experiment



Control Group

The group that gets NO change.

They are the baseline. Everything else is compared to them.

Example: The skateboard with the wheels you'd normally use.



Experimental Group(s)

The group(s) that get the change — the ONE thing you're testing.

Example: The skateboard with bigger wheels.

If you change more than one thing, you can't tell which change caused the difference.

The Three Kinds of Variables



INDEPENDENT

What YOU change.

(I Change)

Only ONE per experiment.



DEPENDENT

What you MEASURE.

The outcome that DEPENDS on your IV.

Only ONE per experiment.



CONTROL

What you KEEP THE SAME.

Everything except the IV. If you don't hold them constant, your data is meaningless.

Practice — Identify the Variables

Does wheel size affect how fast a skateboard travels down a ramp?

IV

Wheel size

What you're changing

DV

Speed

What you're measuring

CV

Same ramp, same skateboard, same rider, same push force

What stays constant

Step 4 — Collect Data



Record Honestly

Write down every measurement — even the ones that look 'weird'.

Outliers can be:

- Real (something to explain)
- A measurement mistake

Either way, you can't decide without the data.



Repeat Trials

One trial is a story.

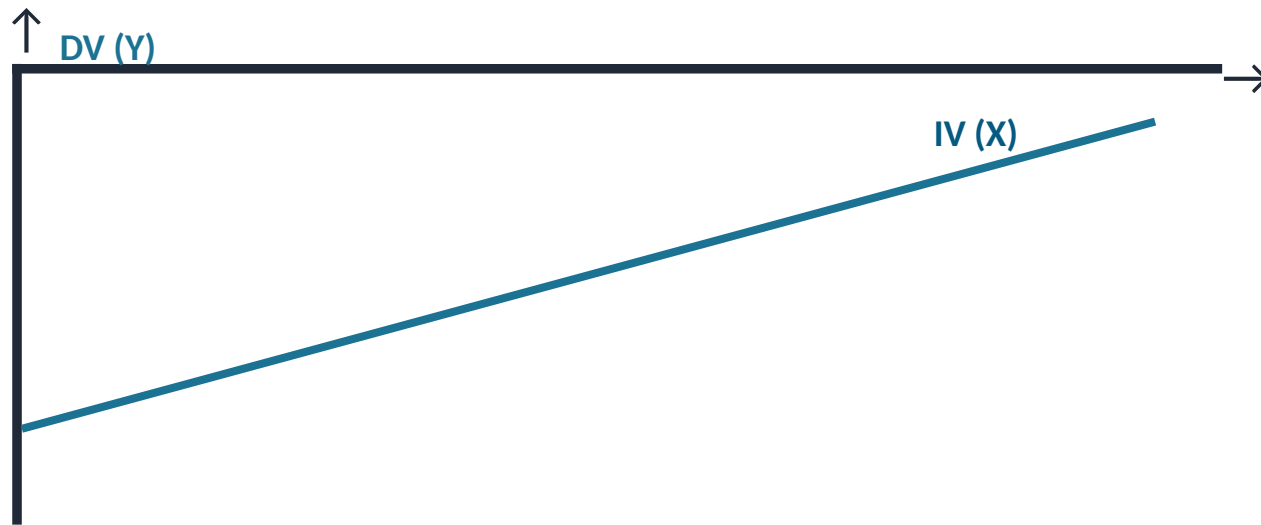
Five trials is EVIDENCE.

Multiple trials let averaging cancel out random variation. This is why we always repeat.

Step 5 — Analyze the Data

Golden Rule of Graphing

Independent Variable → X-axis Dependent Variable → Y-axis



Look for a pattern: "As A increases, B _____"

Step 6 — Conclusion



Report Honestly

Answer:

- Did the data support your hypothesis?
- What is the relationship you found?
- Any surprises?

A good report lets someone else redo your experiment.



What If You're Wrong?

Do NOT go back and change your hypothesis.

A disproved hypothesis is not WRONG — it's DATA. Scientists learn as much from a no as from a yes.

(This is the honest habit that separates science from opinion.)

Full Example — The Broken Flashlight

1

QUESTION

Why is this flashlight not working?

2

HYPOTHESIS

If I replace the batteries, then the flashlight will work.

3

EXPERIMENT

Change the batteries. Turn the flashlight on / off.

4

COLLECT DATA

Does the light now come on?

5

ANALYZE

Yes — the light comes on when batteries are fresh.
No — light stays off, hypothesis disproved.

6

CONCLUSION

If YES: bad batteries were the cause.
If NO: form a new hypothesis (broken bulb? bad switch?).

In Summary

1

Cause & Effect

Ask 'Does A have an effect on B?' — not 'Which is better?'

2

Measurable Hypothesis

Include a specific prediction about the IV/DV relationship.

3

One Variable at a Time

Only change the IV. Everything else stays constant.

4

Graph It

Data table → graph. IV on X, DV on Y.

5

Describe the Relationship

"As A increases, B _____. " Be honest — even if 'B stays the same.'

PRACTICE

Identifying Variables

Work through the online practice on Google Classroom.
For each scenario: identify IV, DV, and CVs, then write an if-then hypothesis.

Bring your answers to class — we'll compare and discuss.